CSE/ISyE 6740
Computational Data Analysis

Diffusion Models

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Outline

- **Advanced Topics**
  - Diffusion Models
    - DDPM loss
    - Diffusion model architecture
    - Computation cost
    - Diffusion model applications

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DDPM Training



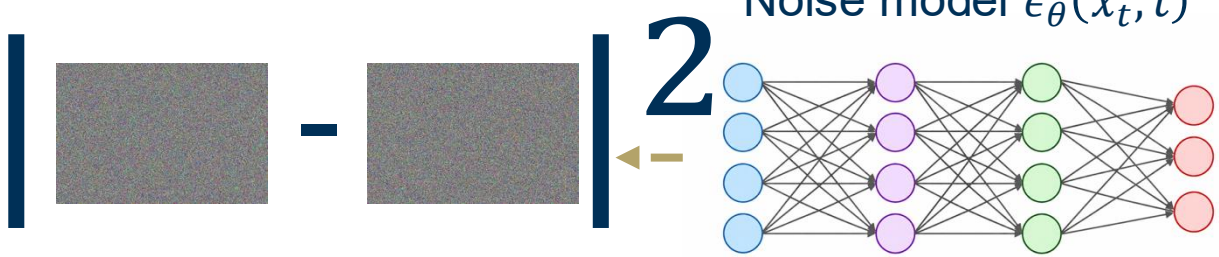
Sample one



Add noise



Noise model $\epsilon_\theta(x_t, t)$



Backpropagate to minimize the loss

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on

$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$

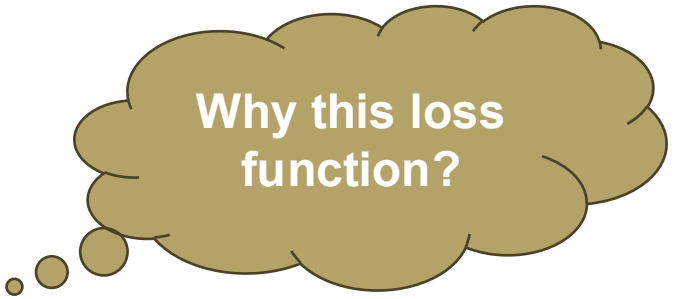
6: **until** converged

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DDPM Algorithm

Denoising Diffusion Probabilistic Models

The loss function can be written as:



$$L_{DDPM} = \sum_t \mathbb{E}_{t\sim[1,T],\, x_0\sim D_{train},\, \epsilon_t\sim\mathcal{N}(0,1)} \left[\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2\right]$$

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

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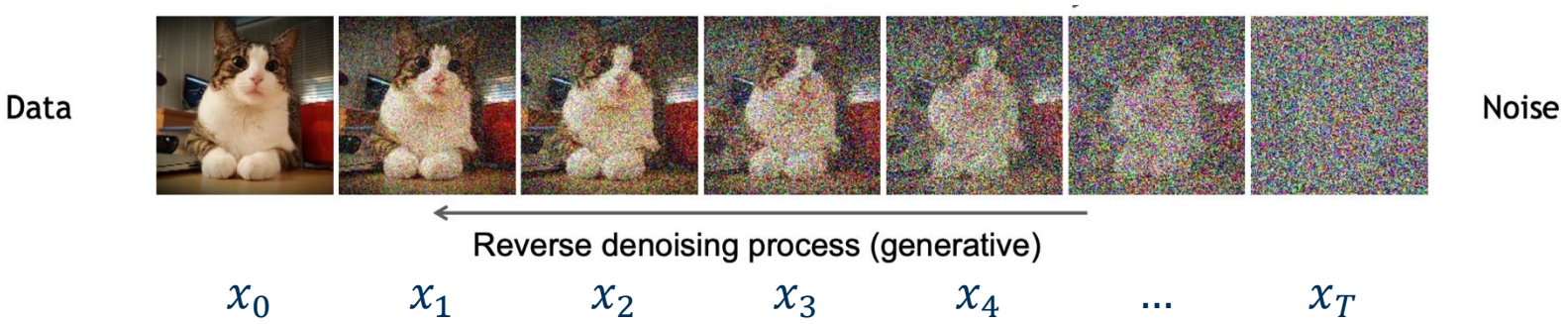
DDPM Loss

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Maximum Likelihood Estimation

- Maximize the likelihood of seeing the same output $x' = x_0$ as input $x = x_0$
- Or minimize negative log-likelihood

$$\min_{\theta} -\mathbb{E}_{x_0 \sim D_{train}}[\log p_\theta(x_0)]$$



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Backpropagation Through Time (BPTT)

Algorithm 2 Sampling

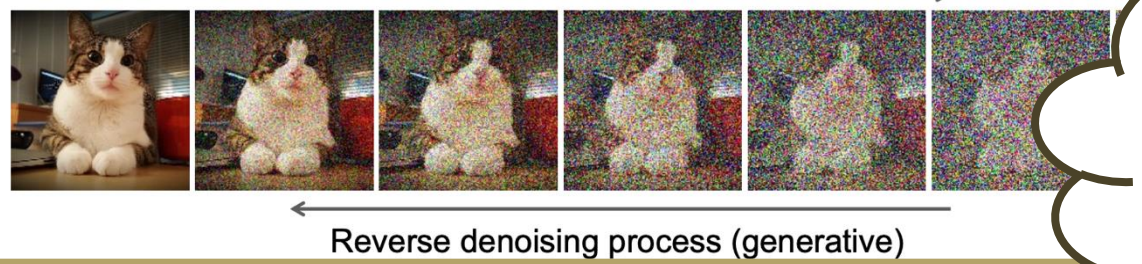
1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

$$\min_\theta -\mathbb{E}_{x_0 \sim D_{train}}[\log p_\theta(x_0)]$$

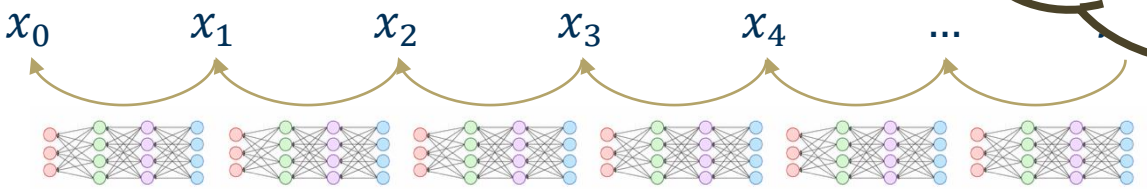
- Backpropagate through the sampling process to compute gradient

$$\frac{d}{d\theta}\log p_\theta(x_0) = \frac{\partial}{\partial\theta}\log p_\theta(x_0) + \frac{\partial \log p_\theta(x_0)}{\partial x_0}\frac{dx_0}{d\theta}$$

$$\frac{dx_0}{d\theta} = \frac{\partial x_0}{\partial x_1} + \frac{\partial x_0}{\partial x_1}\frac{dx_1}{d\theta}, \qquad \frac{dx_1}{d\theta} = \dots$$



What is the computation cost? Memory cost?



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DDPM Loss

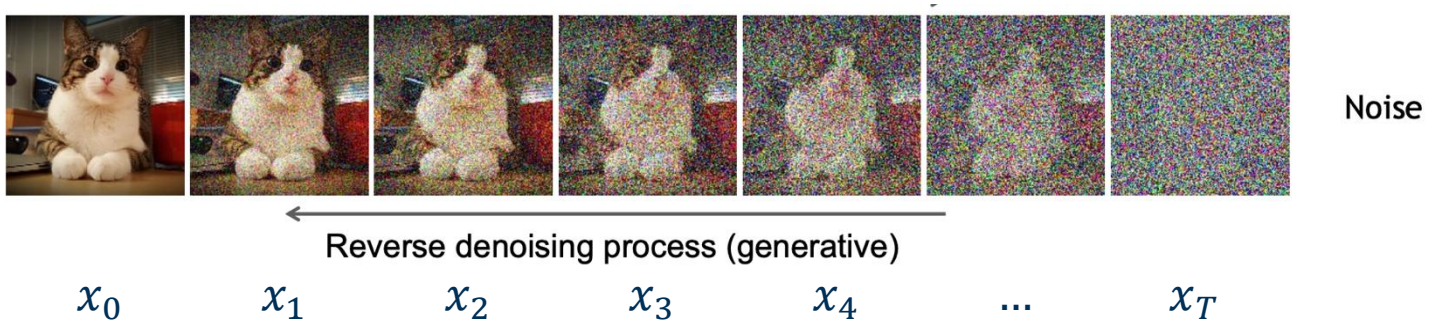
Integrating over any denoising routes $x_{1:T}$

$$-\mathbb{E}_{x_0 \sim D_{train}}[\log p_\theta(x_0)] = -\mathbb{E}_{x_0 \sim D_{train}}\left[\log \int p_\theta(x_{0:T})\, dx_{1:T}\right]$$

$$= -\mathbb{E}_{x_0 \sim D_{train}}\left[\log \int q(x_{1:T}|x_0) \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\, dx_{1:T}\right]$$

$$\leq -\mathbb{E}_{x_0 \sim D_{train},\ x_{1:T} \sim q(x_{1:T}|x_0)}\left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]$$

Jensen inequality



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DDPM Loss

$$L := -\mathbb{E}_{x_0 \sim D_{train},\ x_{1:T} \sim q(x_{1:T}|x_0)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)} \right]$$

$$= \mathbb{E}_{x_0 \sim D_{train},\ x_{1:T} \sim q(x_{1:T}|x_0)} \left[\log \frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})} \right]$$

$$= \mathbb{E}_{x_0 \sim D_{train},\ x_{1:T} \sim q(x_{1:T}|x_0)} \left[\log \frac{q(x_1|x_0) \cdot q(x_2|x_1) \cdot \ldots \cdot q(x_T|x_{T-1})}{p_\theta(x_0|x_1) \cdot p_\theta(x_1|x_2) \cdot \ldots \cdot p_\theta(x_{T-1}|x_T) \cdot p_\theta(x_T)} \right]$$

$$= \mathbb{E}_{x_0 \sim D_{train},\ x_{1:T} \sim q(x_{1:T}|x_0)} \left[-\log p_\theta(x_T) + \sum_{t=1}^{T} \log \frac{q(x_t|x_{t-1})}{p_\theta(x_{t-1}|x_t)} \right]$$

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DDPM Loss

$$\mathbb{E}_{x_0 \sim D_{train},\ x_{1:T} \sim q(x_{1:T}|x_0)} \left[-\log p_\theta(x_T) + \sum_{t=1}^{T} \log \frac{q(x_t|x_{t-1})}{p_\theta(x_{t-1}|x_t)} \right]$$

$$= \mathbb{E}_{x_0 \sim D_{train},\ x_{1:T} \sim q(x_{1:T}|x_0)} \left[-\log p_\theta(x_T) + \sum_{t=1}^{T} \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} \cdot \underbrace{\frac{q(x_t|x_0)}{q(x_{t-1}|x_0)}}_{\text{Bayes rule}} \right]$$

$$= \mathbb{E}_{x_0,\, x_{1:T} \sim q} \left[-\log p_\theta(x_T) + \log q(x_T|x_0) + \sum_{t=1}^{T} \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} \right]$$

$$= \mathbb{E}_{x_0,\, x_{1:T} \sim q} \left[\underbrace{D_{KL}\big(q(x_T|x_0) || p_\theta(x_T)\big)}_{L_T} + \sum_{t=1}^{T} \underbrace{D_{KL}\big(q(x_{t-1}|x_t, x_0) \,||\, p_\theta(x_{t-1}|x_t)\big)}_{L_{t-1}} \right]$$

L_T = constant

L_{t-1} (L_0 is often separated and ignored)

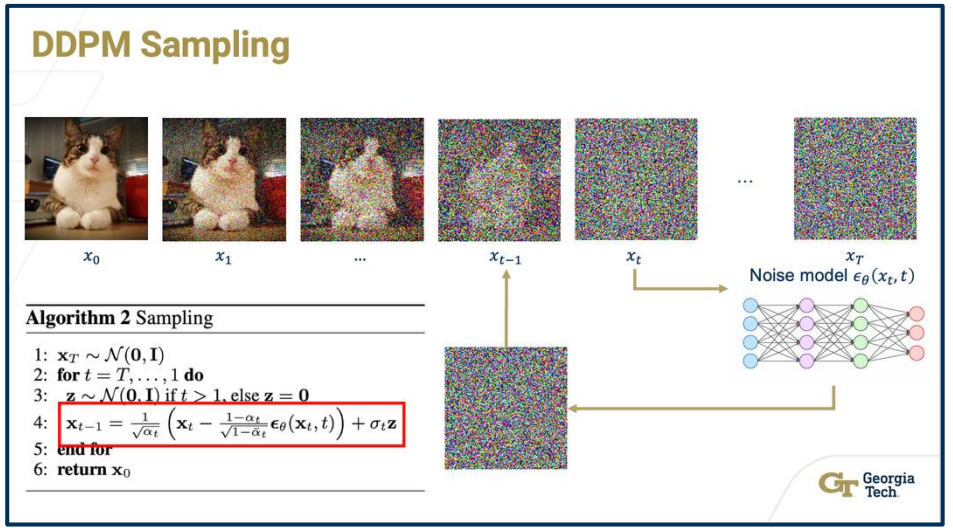
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DDPM Loss Derivation

- What is $L_{t-1} = D_{KL}\left(q(x_{t-1}|x_t, x_0) \,||\, p_\theta(x_{t-1}|x_t)\right)$?

$$= \left\| \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right) - \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) \right\|^2$$

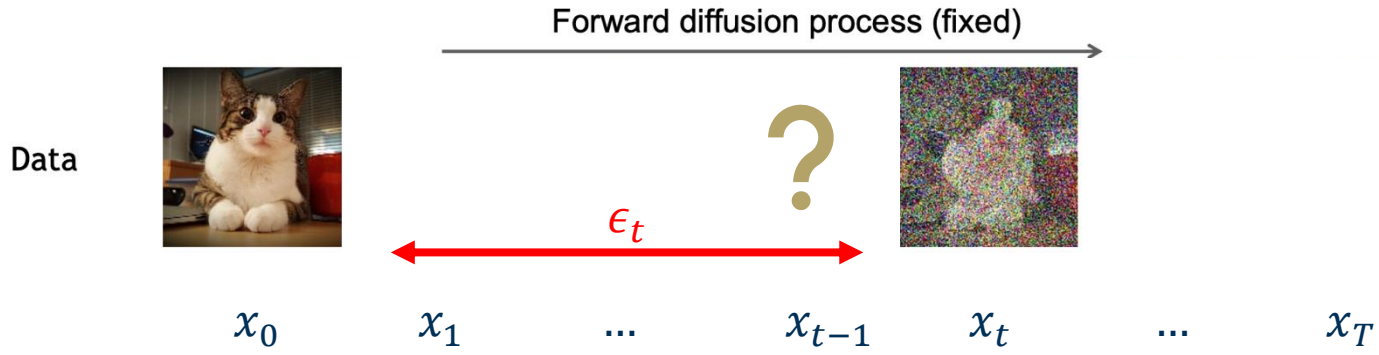
$$= \frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\sigma_t^2}\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2$$



$p_\theta(x_{t-1}|x_t)$ is a Gaussian with mean $\mu_\theta(x_{t-1}|x_t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right)$ and variance σ_t^2

$q(x_{t-1}|x_t, x_0)$ is a Gaussian with mean?

- $q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)}$ with mean $\frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right)$



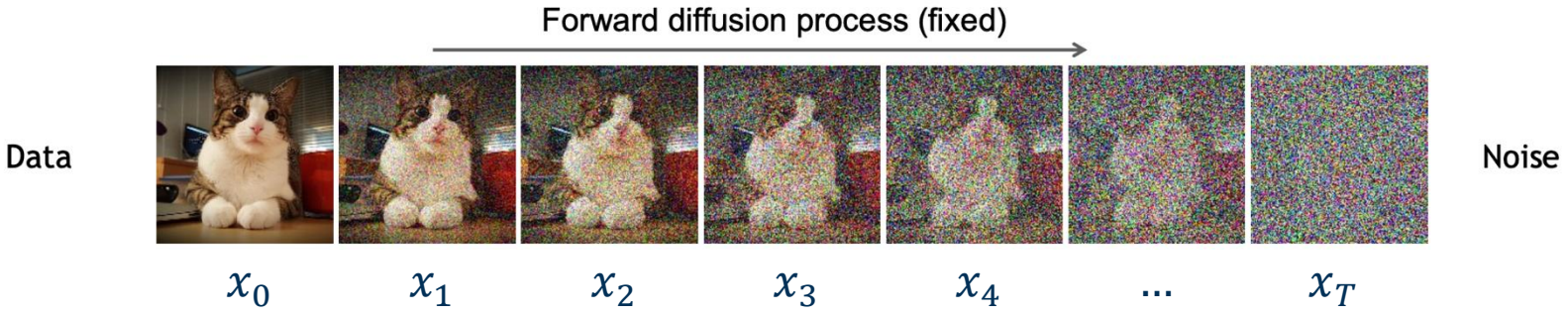
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Learn the Noise Model

- Train a **noise model** $\epsilon_\theta(x_t, t)$ to predict how much noise to deduct from the corrupted image/sample x_t.

$$-\log p_\theta(x_0) \leq L = \mathbb{E}\left[\sum_{t=1}^{T-1} L_t\right] = \mathbb{E}\left[\sum_{t=1}^{T-1} \frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\sigma_t^2} \|\epsilon_t - \epsilon_\theta(x_t, t)\|^2\right]$$

$$L_{DDPM} := \mathbb{E}_{t\sim[1,T],\, x_0\sim D_{train},\, \epsilon_t\sim\mathcal{N}(0,1)} \left[\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2\right]$$



Please read https://lilianweng.github.io/posts/2021-07-11-diffusion-models/ to understand how the DDPM derivation works and finish your homework 3.

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Demos

Signal multiplier: 0.9203 · noise multiplier: 0.3911

This image's noise MSE: 0.035762 · mean over all 3 images at this t: 0.030236

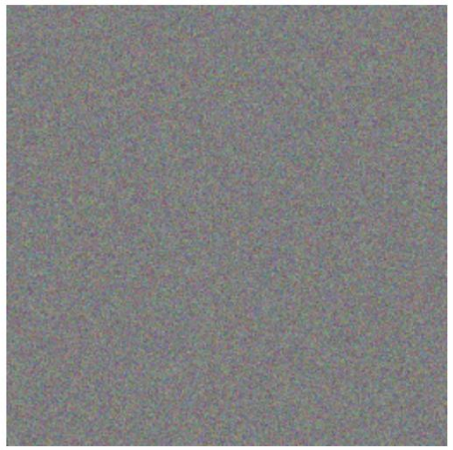
1. Reference image x_0



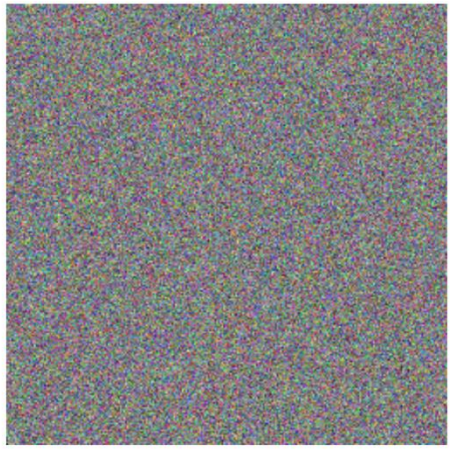
2. Noisy input x_t



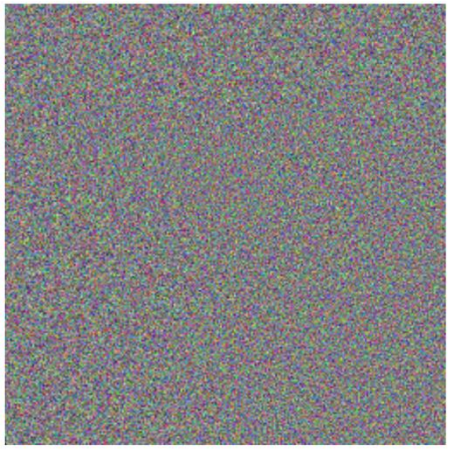
3. Scaled noise added



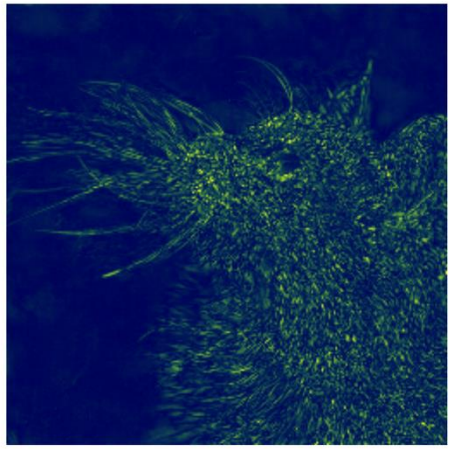
4. True noise ε (target)



5. Predicted noise εθ



6. Squared error heatmap



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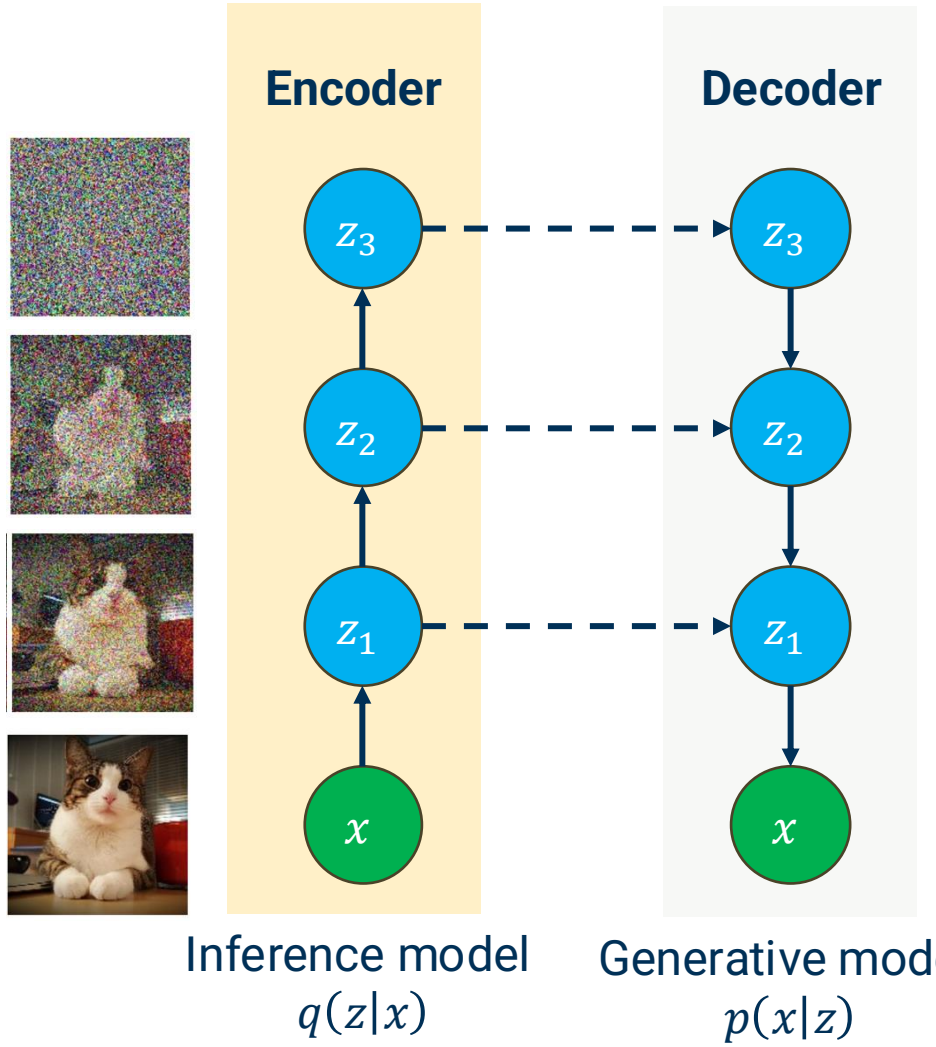
Diffusion Model Architecture

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Connection to VAE

- Diffusion models can be considered as a special form of **hierarchical VAEs**

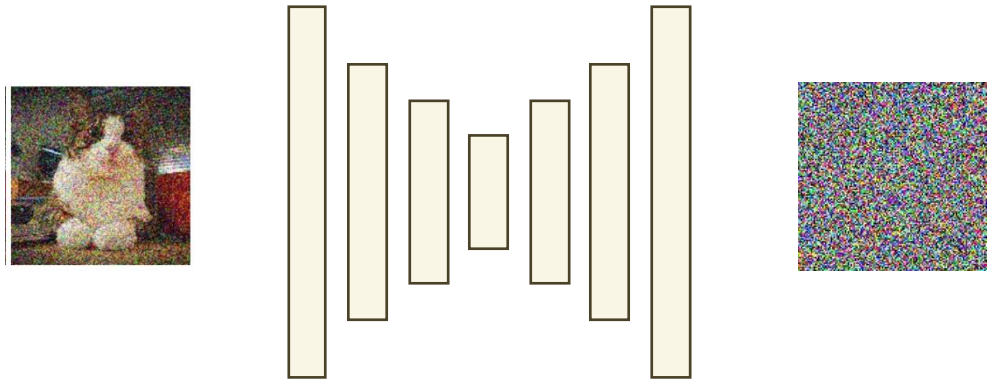
- In diffusion models:
  - The encoder is fixed
  - The latent variables have the same dimension as the data
  - The loss is decomposed to each time step
  - The model is trained with a weighted ELBO



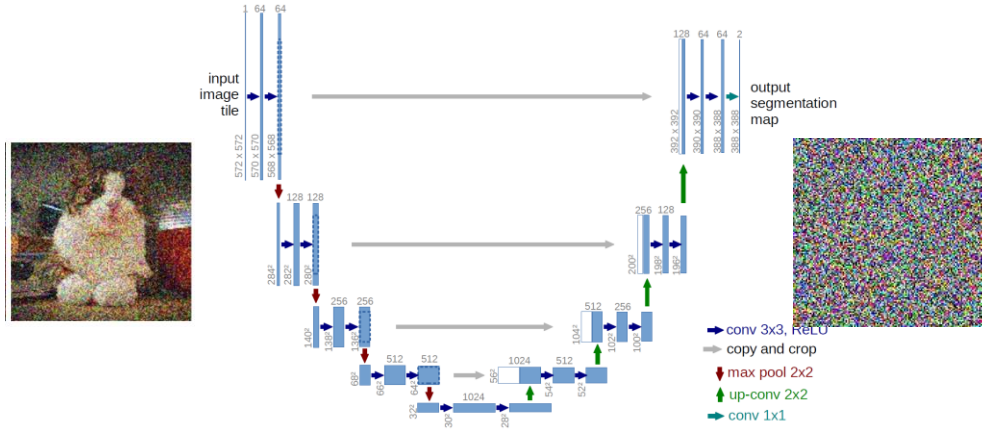
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Model Architecture

- **Design criterion**: Input and output must have the same size



Fully connected layer



U-Net for images
(Conv2D + ConvTranspose2D)

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Time Representation

- The noise model $\epsilon_\theta(x_t, t)$ consumes
  - **Noisy input x_t**
  - **Time representation of t**: different diffusion time step may have different noise distribution

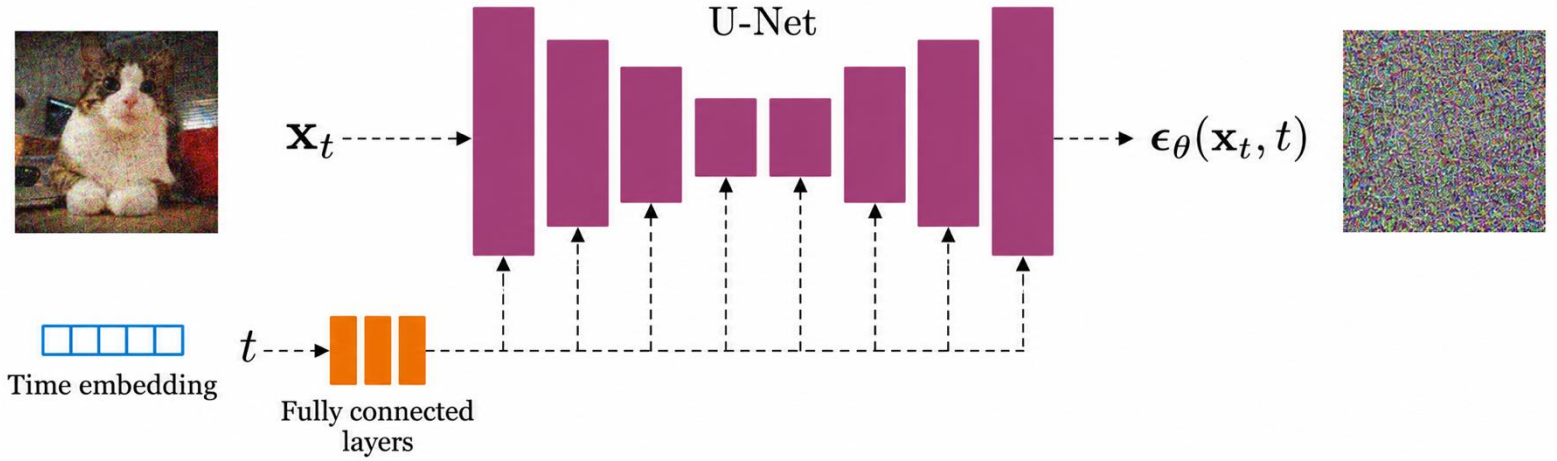


Image credit https://cvpr2022-tutorial-diffusion-models.github.io/

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Conditional Diffusion Model

- The noise model $\epsilon_\theta(x_t, t)$ consumes
  - **Noisy input x_t**
  - **Time representation of t**: different diffusion time step may have different noise distribution
  - **Class/context y**

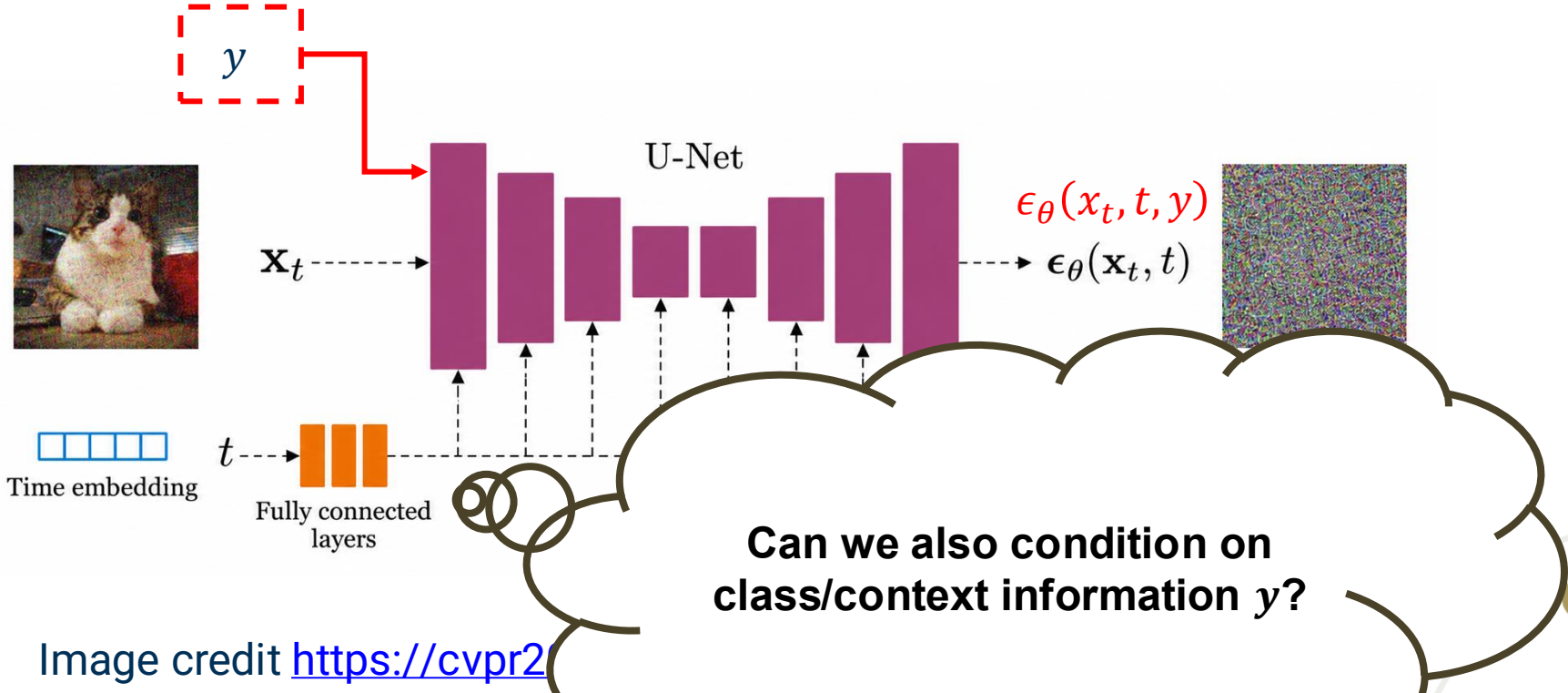


Image credit https://cvpr2

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Training Demo

- Computer spec: MacBook Pro with an Apple M3 Pro
  - One 12-core CPU
  - 18-core GPU
  - 36 GB of unified memory

- Image size: 256 x 256 x 3
- Sampling steps: $T = 200$
- Noise model: U-Net with Conv2D x3 + ConvTranspose2D x3

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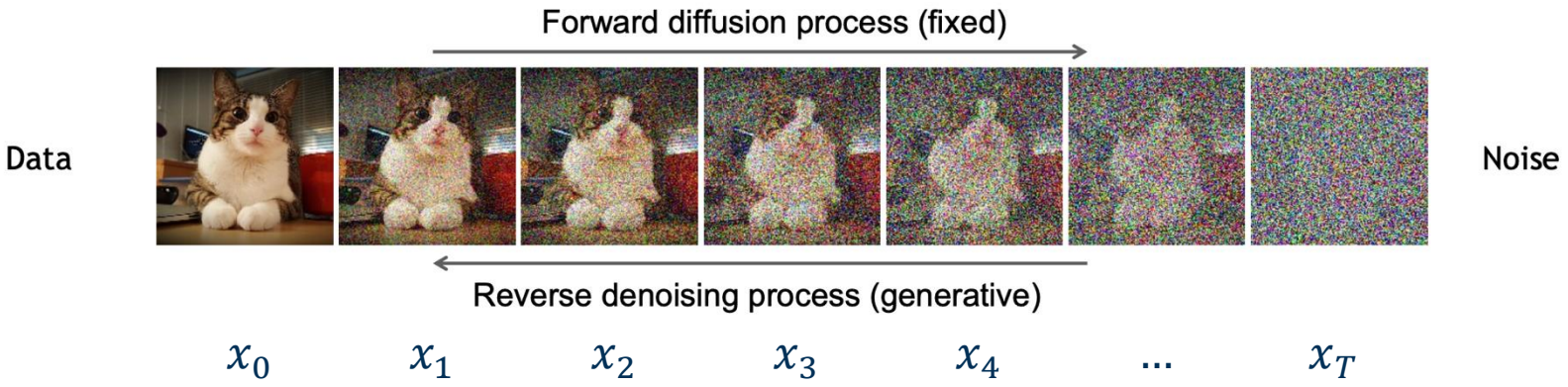
Computation Cost

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Training Cost

- What is the training cost per gradient iteration for diffusion model $\epsilon_\theta(x, t)$?

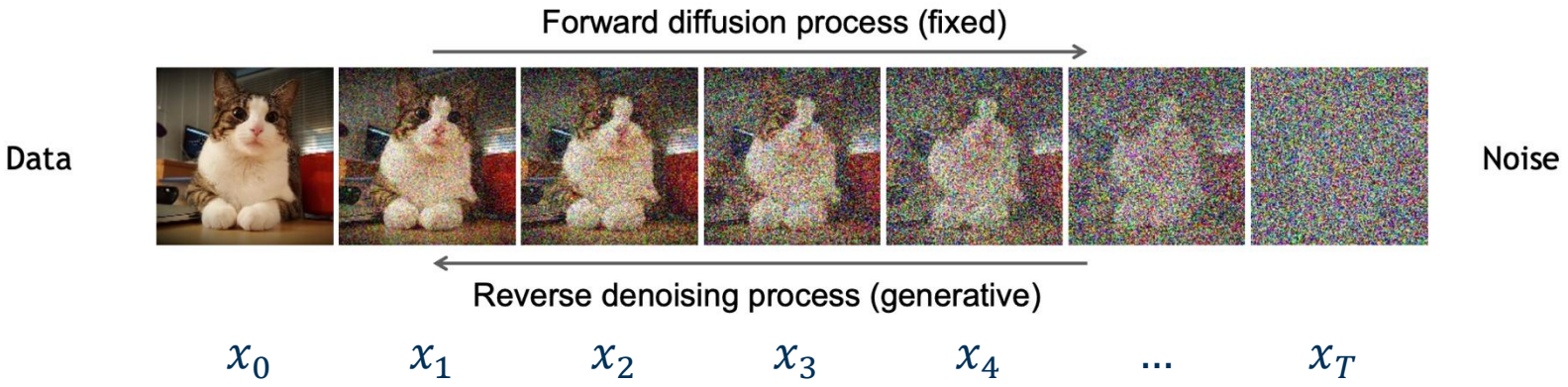
$$L_{DDPM} \coloneqq \mathbb{E}_{t\sim[1,T],\, x_0\sim D_{train},\, \epsilon_t\sim\mathcal{N}(0,1)} \left[\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2\right]$$



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Sampling Cost

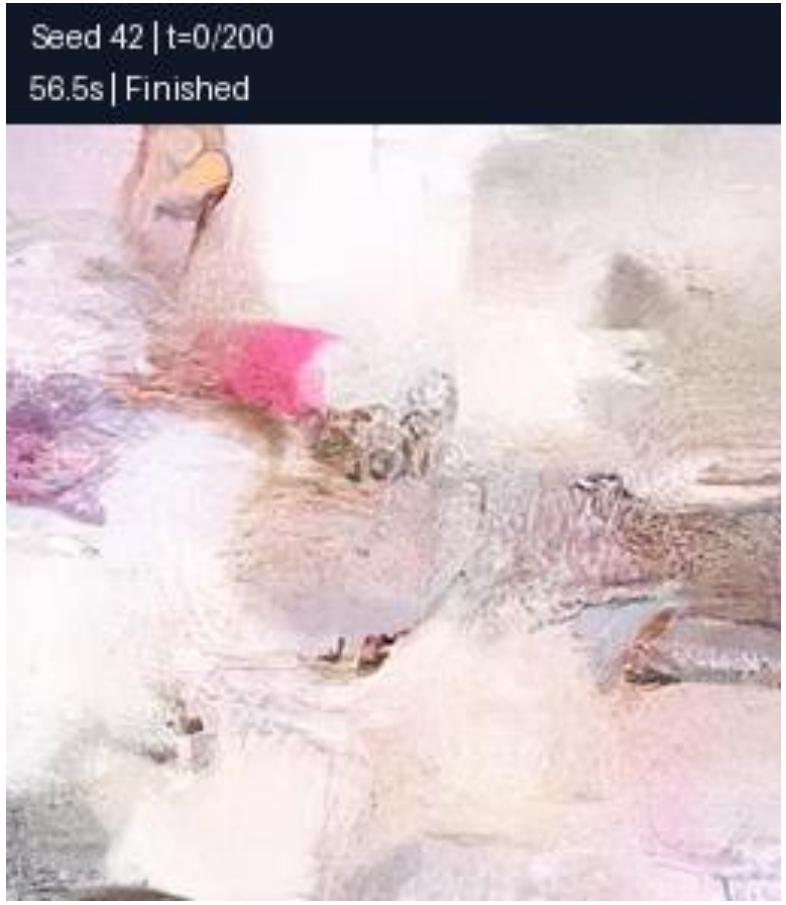
- What is the sampling cost per gradient iteration for diffusion model $\epsilon_\theta(x, t)$?



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Sampling Demos





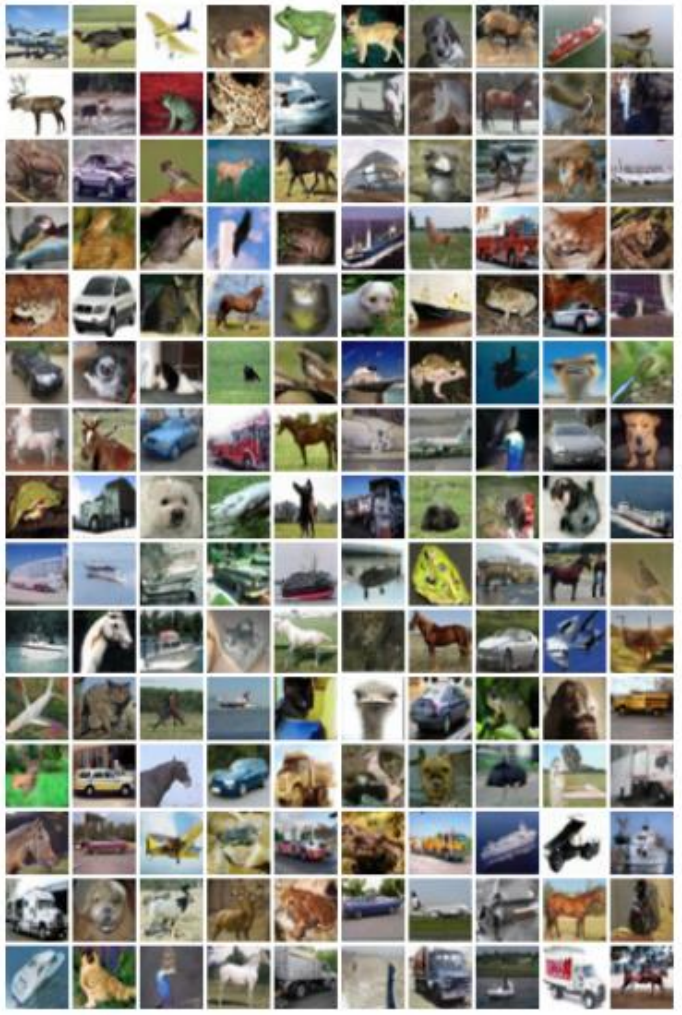
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Diffusion Model Application

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Diffusion Model Applications





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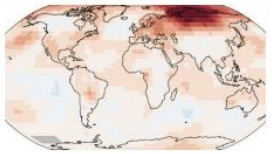
Applications of VAE and Diffusion Models?

- We see so many fancy applications of VAE and diffusion models. But can we use them to do something else?
- Technically, these models are trying to do **density estimation**
  - Try to estimate $p(x)$ and $p(x \mid y)$ from a dataset D
- We have parametric models (Gaussian, GMM etc), but VAE and diffusion models offer a **deep learning option** for us.

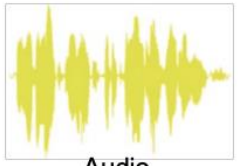

EEG (Healthcare)


Traffic


Recommendation


Climate and Weather


Energy and Electricity


Audio


AIOps

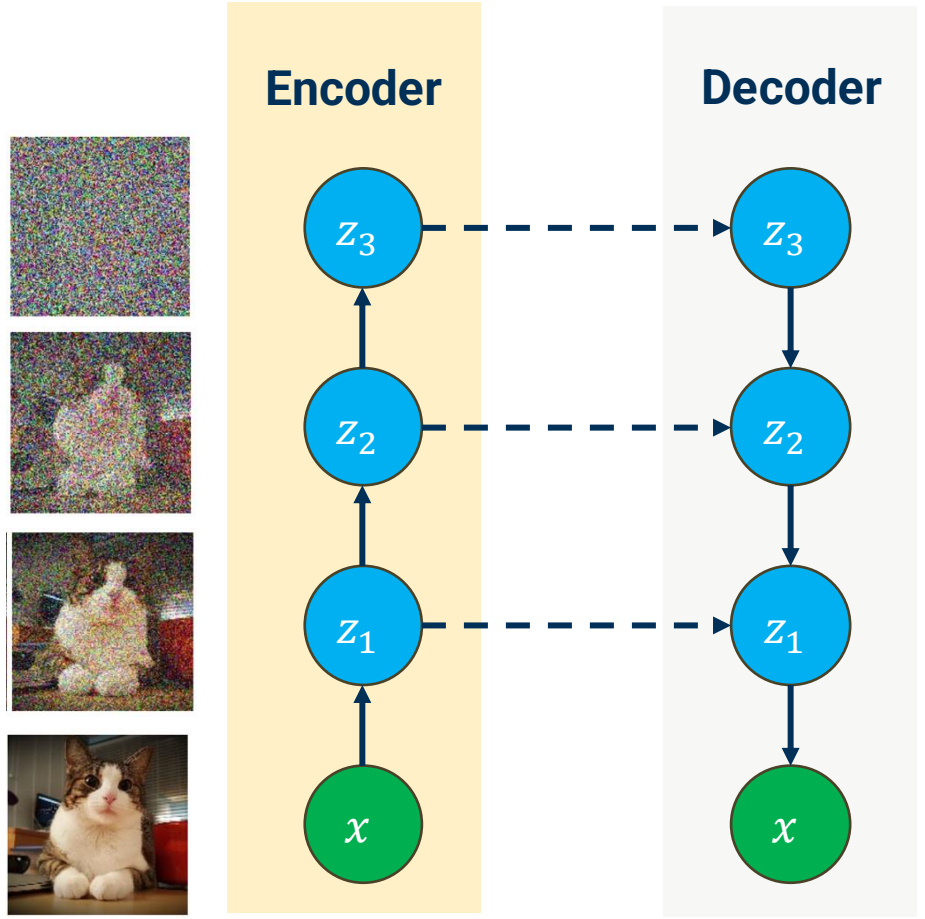

Finance

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Comparison of VAEs and Diffusion Models

- **VAE**
  - 👑 Stronger latent space representation
  - 👑 Can be used to obtain embedding
  - 👑 Less compute needed
  - More unstable
  - Lower resolution

- **Diffusion models**
  - Simpler latent space representation
  - No encoder learned
  - More compute needed
  - 👑 More stable
  - 👑 Higher resolution



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References

- “What are Diffusion Models?” by Lilian Wang
- https://cs231n.stanford.edu/slides/2023/lecture_15.pdf
- CVPR 2022 tutorial: “Denoising Diffusion-based Generative Modeling: Foundations and Applications”
- “Denoising Diffusion Probabilistic Models”, Jonathan Ho, Ajay Jain, Pieter Abbeel 2022

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